



# VISION

## The Future of High-Integrity AI Assessments

Secure Exam Browser & Real-Time Proctoring Platform — built by **Vinit Jangir & Adarsh**

# The Trust Crisis in Remote Exams

## Impersonation

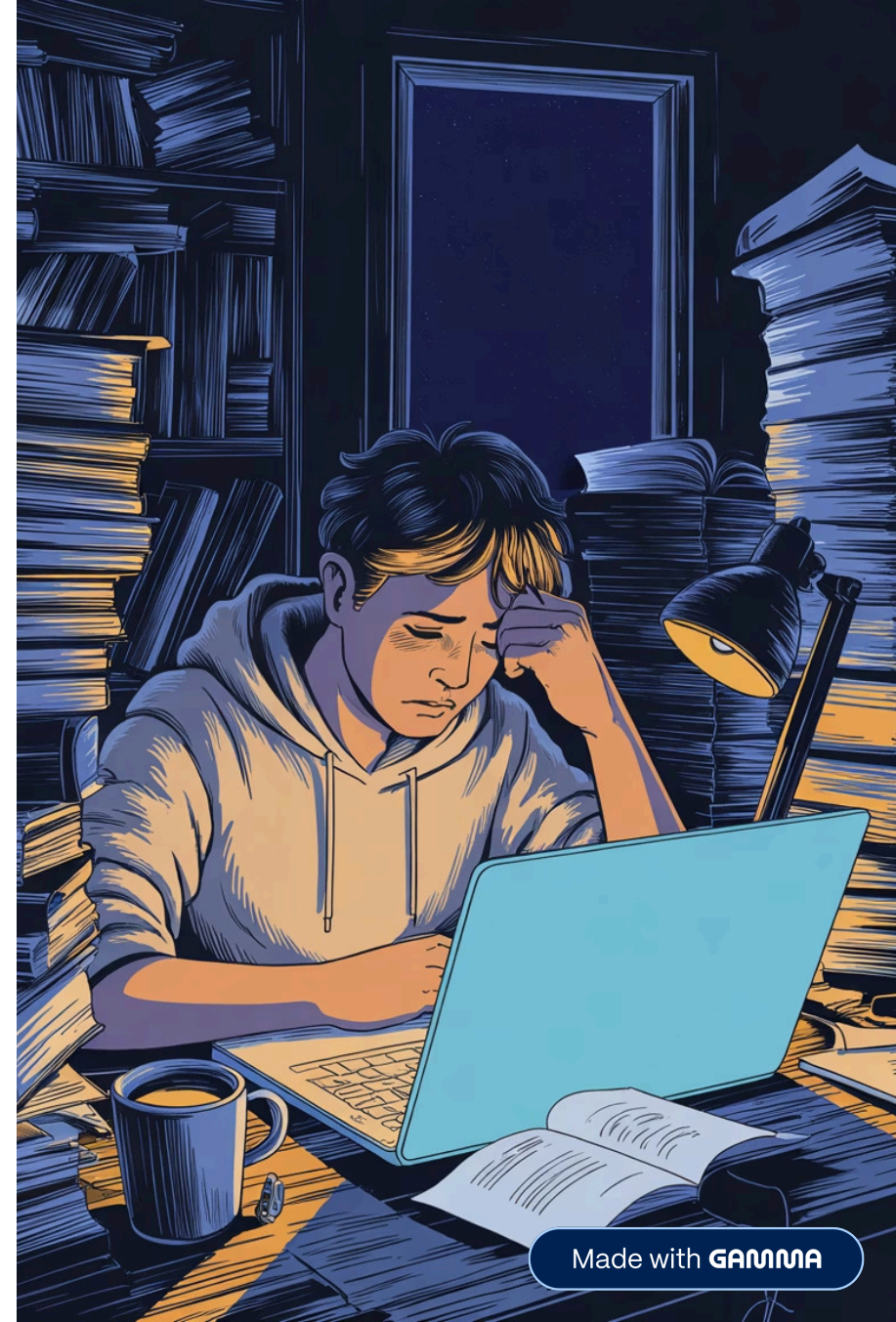
Unverified identities allow proxy test-takers to fraudulently complete exams on behalf of enrolled students.

## Academic Dishonesty

Tab-switching, screen sharing, and collusion during live assessments undermine institutional integrity.

## Grading Overload

Manual evaluation of hundreds of code submissions creates bottlenecks and delays result delivery by days.



# A Triple-Layered Defense

VISION addresses the trust problem with three interlocking systems — each engineered to close a specific vulnerability in the remote exam lifecycle.



## eKYC Identity Gate

Selfie + Government ID capture with mandatory admin approval before exam access is granted.



## Anti-Cheat Engine

Hardened browser with tab-switch detection, fullscreen enforcement, and real-time webcam monitoring.



## Grading Pipeline

Asynchronous BullMQ queue with Judge0 execution delivers instant, automated code evaluation at scale.





#### FEATURE SPOTLIGHT

# eKYC & Identity Verification

No candidate enters the exam room without passing VISION's identity gate — eliminating impersonation at the source.

## Selfie + ID Capture

Mandatory dual-factor verification: a live selfie paired with a government-issued photo ID is submitted before exam start.

## Admin Approval Board

Proctors review a live dashboard of pending verifications, approving or rejecting candidates in real time before unlock.

# The Anti-Cheat Engine

A hardened browser environment transforms any device into a locked-down exam terminal — with zero tolerance for suspicious behavior.



## Tab-Switch Detection

Any attempt to leave the exam window is flagged instantly, logged to the proctor dashboard, and counts as a violation strike.



## Real-Time Webcam Snapshots

Periodic automated snapshots are captured throughout the exam and surfaced to proctors for asynchronous review.



## Fullscreen Enforcement

Exiting fullscreen mode triggers a real-time alert. Repeated violations can auto-submit and terminate the session.



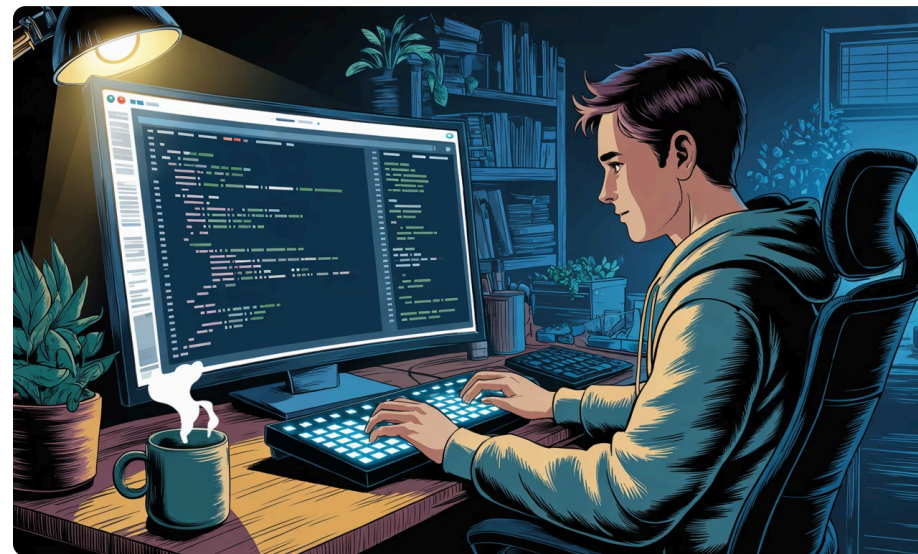
## Copy-Paste & DevTools Block

Keyboard shortcuts, right-click context menus, and developer tools are disabled system-wide during active sessions.

INNOVATION

# The LeetCode Question Scraper

VISION eliminates the friction of manual question entry with a proprietary scraping engine built on **Puppeteer**.



## 1 Paste a URL

Mentors submit any LeetCode problem URL directly inside the VISION admin console.

## 2 Puppeteer Parses It

The headless browser engine extracts the problem title, description, constraints, and examples automatically.

## 3 Instant Question Bank

Structured data is saved to the exam question bank — ready to assign in seconds, no reformatting required.

# Technical Excellence

Built for high-concurrency enterprise environments — every layer of the VISION stack is chosen for performance, reliability, and scale.



## React

Responsive, component-driven frontend for candidate and admin interfaces.



## Node.js

Non-blocking API server handling concurrent exam sessions at scale.



## Socket.IO

Bi-directional real-time channel for live proctor alerts and session events.



## Redis

In-memory rate limiting and session state management for sub-millisecond lookups.



## BullMQ + Judge0

Async job queue paired with a sandboxed code execution engine for safe, scalable grading.

# Roadmap & Vision Ahead

1

## Phase 1 — Now

eKYC gate, hardened browser, LeetCode scraper, and async grading pipeline — fully operational.

2

## Phase 2 — Next

AI-driven answer evaluation with NLP scoring models for subjective and short-answer question types.

3

## Phase 3 — Future

Face recognition continuous authentication, LMS integrations (Canvas, Moodle), and enterprise SSO support.

"Built for Integrity. Scaled for Excellence."

